

Erdős Problem #336

EP-336

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

For $r \geq 2$ let $h(r)$ be the maximal finite k such that there exists a basis $A \subseteq \mathbb{N}$ of order r (so every large integer is the sum of at most r integers from A) and exact order k (so every large integer is the sum of exactly k integers from A).

Find the value of $\lim_r \frac{h(r)}{r^2}$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>